LARGE BIPARTITE GRAPHS OF GIVEN DEGREE AND DIAMETER FROM GENERALIZED POLYGONS

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ABSTRACT. Let $n_b(\Delta, D)$ be the largest order of a bipartite graph of maximum degree at most Δ and diameter at most D. We give constructions from generalized polygons that improve the lower bounds in the 2013 table of Miller and Širáň at diameters 3, 4, 6 and 7, and Haar graphs that improve it at diameters 5, 8, 9 and 10. At diameter 3 we add a fourth hat to the three hats of Araujo-Pardo and López and obtain $n_b(q+2,3) \geq 2(q^2+q+6)$ with regular graphs. At diameter 4 we copy points and lines of a generalized quadrangle with weights and stand-ins; a weighted 2-ovoid of the Ahrens–Szekeres quadrangle built from two elliptic quadrics gives $n_b(q+2,4) \geq 2q^2(q+2)$ for odd q, and Singer 2-ovoids of $W(3,q)$ give $n_b(q+3,4) \geq 2(q^2+1)(q+3)$, both with regular graphs. At diameter 6 an explicit set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$ ("whiskers") added to Delorme's doubling gives $n_b(q+m,6) \geq 2(q+1)(q^4+q^2+1)+(m-1)\big(2q^4+2q^3+2q+2+m(q^3+q+2)+m^2\big)$, and a regular version. At diameter 7 we join $2r+1$ copies of the incidence graph of $H(q)$ by matchings, delete matching edges at suitable sets and attach new vertices; we prove which sets are allowed. All constructions are given in projective coordinates with the exact neighbourhood of every new vertex. We also show that the graphs of Gómez behind six diameter-8 entries of the table have diameter at least 9 as defined, and that relabelling their copies by automorphisms does not help.

POINTS FOR THE CO-AUTHORS

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- AI disclosure. The constructions, proofs and checks in this draft were produced in a research repository run by Claude Code agents coordinated by D. Özden, and most explicit ingredients (sets, connection sets, hub graphs) were found by computer search. Following the guidance of the 20 October meeting this draft keeps search and AI out of the mathematics and mentions computation only as "computer-checked". Decide how to acknowledge the AI use (most journals now ask for a statement) and whether to cite the repository `https://github.com/dogaozden/degree-diameter`, which holds every graph, builder and check log. 2
- Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16,3)$ 396; $(4,5)$ 162, $(5,5)$ 354, $(8,5)$ 2 048, $(9,5)$ 2 758, $(10,5)$ 4 800, $(11,5)$ 7 020, $(12,5)$ 9 840, $(13,5)$ 11 700, $(14,5)$ 17 568, $(15,5)$ 20 736, $(16,5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4,7)$ 864, $(5,7)$ 3 120, $(6,7)$ 8 880, $(8,7)$ 43 344; $(3,10)$ 810; and later Haar graphs at $(9,8)$ 546 720, $(5,9)$ 29 756, $(7,10)$ 2 298 240, $(8,10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper. 4

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The referee of the diameter-7 note recalls that this form of the quadric and of Tits' equations is [?, §2.4.13]. We could not consult [VM98] to confirm the place; please check and cite it. 5

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1. Introduction

A bipartite graph of maximum degree $\Delta \geq 2$ and diameter $D \geq 2$ has at most

$$M_b(\Delta, D) = 2\big(1 + (\Delta - 1) + \cdots + (\Delta - 1)^{D-1}\big)$$

vertices (the bipartite Moore bound). Let $n_b(\Delta, D)$ be the largest order of such a graph. For $D \geq 3$ and $\Delta \geq 3$ the bound is attained only by the incidence graphs of generalized polygons: projective planes ($D = 3$), generalized quadrangles ($D = 4$) and generalized hexagons ($D = 6$) of order $(\Delta - 1, \Delta - 1)$ [?, ?]. These are known when $\Delta - 1$ is a prime power. For the other cells only lower bounds are known. They are collected in Table 4 of the survey of Miller and Širáň [?] (version of 2013), which the CombinatoricsWiki [?] reproduces; since 2013 the wiki has changed only at $(13, 3)$ and $(15, 3)$, by the work of Araujo-Pardo and López [?]. We call this table "the 2013 table".

This paper gives constructions that improve the 2013 table in the cells listed in Table 1. Most of them are built from generalized polygons in their classical coordinates, by three kinds of operations: copying points and lines (with Delorme's doubling with levels [?] as the basic tool), changing a bounded number of incidences, and joining several copies of an incidence graph by matchings and new vertices. The remaining cells are Haar graphs (bi-Cayley graphs) of explicit groups.

Results.

- *Diameter 3* (Section **??**). Araujo-Pardo and López [?] added three "hats" to $\mathrm{PG}(2, q)$ and obtained $n_b(q + 2, 3) \geq 2(q^2 + q + 5)$. We add a fourth hat (Theorem **??**): $n_b(q + 2, 3) \geq 2(q^2+q+6)$ for every prime power $q \geq 4$, attained by a $(q+2)$-regular graph; the construction works in every projective plane of order $q \geq 3$.
- *Diameter 4* (Section **??**). Theorem **??**: in any generalized quadrangle, clones of points with arbitrary weights and copies of lines with "stand-ins" keep diameter 4. With a weighted 2-ovoid of the Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ made of two elliptic quadrics (Theorem **??**) this gives $n_b(q + 2, 4) \geq 2q^2(q + 2)$ for odd q; with the Singer 2-ovoids of $W(3, q)$ (Proposition **??**) it gives $n_b(q + 3, 4) \geq 2(q^2 + 1)(q + 3)$ for odd q; both with regular graphs. For every prime power $q \leq \Delta - 1$, $n_b(\Delta, 4) \geq 2q^2\Delta$ (Corollary **??**).

- *Diameter 6* (Section **??**). A set of $q^2(q^2-1)$ lines of the split Cayley hexagon $H(q)$, given by an explicit rule (Theorem **??**), can be added at level 0 to Delorme's doubling of $H(q)$ along a Hermitian spread. This gives Theorem **??**, and deleting repeated lines and completing gives regular graphs (Theorem **??**).
- *Diameter 7* (Section **??**). The 2013 values at $(7,7)$ and $(9,7)$–$(16,7)$ are the orders $(2r+1)\cdot 2|H(q)|$ of $2r+1$ copies of the incidence graph of $H(q)$ joined by matchings. We delete matching edges at suitable sets of elements and attach new vertices to the freed slots. Theorem **??** reduces the diameter to one condition per pair of copies; Theorems **??** and **??** give two sufficient conditions (segment sets at automorphic transitions; local dualities), the first also necessary. With classical sets this improves all nine cells (Corollaries **??**–**??**); explicit larger sets give the values of Table 1.
- *Diameter 8* (Section **??**). Six entries of the 2013 table at $D = 8$ are the orders of the compound graphs $\Omega\mathrm{H}_q$ of Gómez [**?**], for which the thesis states diameter at most 8 without proof. As defined there, these graphs have diameter at least 9 (Theorems **??** and **??**), and relabelling the copies by automorphisms of the hexagon does not give diameter 8 (Theorems **??** and **??**).
- *Haar graphs* (Section **??**). We state the covering criterion that decides the diameter of a Haar graph from two breadth-first searches, and list Haar graphs that improve two cells at $D = 5$ and 34 cells at $D = 8, 9, 10$. These are single explicit graphs, checked by computer.

Problem 4 of [**?**] asks to generalize the hat construction to regular bipartite graphs of diameters 4 and 6. The hat lemma (Lemma 2.7) says in what sense it generalizes, and Corollaries **??** and **??** and Theorem **??** give regular graphs at $D = 4$ and $D = 6$ that beat the 2013 table.

What is proved and what is computed. Every theorem in Sections 2–**??** has a complete proof here, with one kind of exception, always marked: a statement about one explicit finite object (a set of points, a connection set, a hub graph, a count of route words) is *computer-checked*, and we name the check. "Computer-checked" for a graph means: built from its definition, and found to be bipartite, of the stated maximum degree and of the stated diameter by breadth-first search from every vertex (or from one vertex of every orbit of an automorphism group that was itself checked), by at least two programs written separately. Appendix **??** lists the checks. Many of the statements are also proved in Lean 4 with Mathlib; Appendix **??** says exactly which.

Table 1: The cells improved in this paper. M_b: bipartite Moore bound; 2013: [**?**, Table 4]; "reg.": the largest regular graph we give, when the record graph is not regular. Values at $D = 8, 9, 10$ are in Table **??**.

(Δ, D)	M_b	2013	this paper	reg.	construction
(11,3)	222	190	**192**		fourth hat, $q = 9$ (Thm. **??**)
(13,3)	314	270 [a]	**276**		fourth hat, $q = 11$
(15,3)	422	370 [a]	**376**		fourth hat, $q = 13$
(7,4)	518	346	**359**	350	clones and shadows on $Q(4,5)$ (Ex. **??**)
(11,4)	2 222	1 734	**1 782**		AS(9), two quadrics (Cor. **??**)
(13,4)	3 770	3 064	**3 146**		AS(11)
(15,4)	5 910	4 946	**5 070**		AS(13)
(16,4)	7 232	5 134	**5 440**		Singer 2-ovoid of $W(3,13)$ (Cor. **??**)
(7,5)	3 110	1 134	**1 152**		Haar graph (Sec. **??**)
(9,5)	9 362	2 496	**2 662**		Haar graph
(7,6)	18 662	8 992	**9 592**	9 348	whiskers in $H(5)$ (Thm. **??**, **??**)
(11,6)	222 222	142 464	**148 944**	147 500	$H(9)$

(Δ, D)	M_b	2013	this paper	reg.	construction
(13,6)	542 906	374 452	**388 972**	386 328	$H(11)$
(15,6)	1 158 390	842 048	**870 440**	866 068	$H(13)$
(16,6)	1 627 232	884 062	**940 846**	927 724	$H(13)$, $m = 3$
(7,7)	111 974	23 436	**23 598**		3 levels of $H(5)$, local duality (Sec. **??**)
(9,7)	599 186	117 648	**118 238**		3 levels of $H(7)$
(10,7)	1 195 742	224 694	**225 152**		3 levels of $H(8)$
(11,7)	2 222 222	398 580	**413 180**		3 levels of $H(9)$, duality
(12,7)	3 897 434	664 300	**693 500**		5 levels of $H(9)$, duality
(13,7)	6 514 874	1 062 936	**1 064 472**		3 levels of $H(11)$
(14,7)	10 458 086	1 771 560	**1 789 588**		5 typed levels of $H(11)$
(15,7)	16 217 462	2 480 184	**2 501 491**		7 typed levels of $H(11)$
(16,7)	24 408 482	4 022 340	**4 052 108**		5 typed levels of $H(13)$

[a] The current values of [**?**] are 274 and 376 [**?**].

All graphs in Table 1 except the four Haar graphs at $D = 5$ and $D = 8$–10 are members of families proved here for every prime power q (or every odd q), with explicit ingredients; at $D = 4$ (7,4) and at $D = 7$ the values in the table use, in addition, explicit sets that are computer-checked, and Section **??** also gives values beating the 2013 table in all nine cells from classical ingredients only (Table **??**).

Scope. The repository also holds graphs at other cells that are not covered by the refereed notes this draft is built from: $(16, 3)$ 396; $(4, 5)$ 162, $(5, 5)$ 354, $(8, 5)$ 2 048, $(9, 5)$ 2 758, $(10, 5)$ 4 800, $(11, 5)$ 7 020, $(12, 5)$ 9 840, $(13, 5)$ 11 700, $(14, 5)$ 17 568, $(15, 5)$ 20 736, $(16, 5)$ 33 792 (lifts of generalized quadrangles and other lifts); $(4, 7)$ 864, $(5, 7)$ 3 120, $(6, 7)$ 8 880, $(8, 7)$ 43 344; $(3, 10)$ 810; and later Haar graphs at $(9, 8)$ 546 720, $(5, 9)$ 29 756, $(7, 10)$ 2 298 240, $(8, 10)$ 8 043 840. Decide whether this paper should include them (they would need write-ups and referee reports first), mention them in a remark, or leave them for a second paper.

Organisation. Section 2 fixes notation, coordinates for the classical polygons, and Delorme's doubling with levels, with a proof and the hat lemma. Sections **??**–**??** treat diameters 3, 4, 6 and 7, Section **??** the Gómez entries at diameter 8, and Section **??** Haar graphs. Section **??** collects the updated values. Appendix **??** writes out the $(7, 4)$ example, Appendix **??** lists what is checked in Lean, and Appendix **??** the computer checks.

2. Preliminaries

2.1. **Incidence structures.** An *incidence structure* $\mathcal{S} = (\mathcal{P}, \mathcal{L}, \mathrm{I})$ has points, lines and a symmetric incidence relation; repeated points and repeated lines (with the same incidences) are allowed. Its *incidence graph* $\Gamma(\mathcal{S})$ has vertex set $\mathcal{P} \cup \mathcal{L}$ and the incident pairs (*flags*) as edges; it is bipartite with the points on one side. A non-incident point–line pair is an *anti-flag*. Every bipartite graph with a chosen 2-colouring is of this form, so

$$n_b(\Delta, D) = \max\{|\mathcal{P}| + |\mathcal{L}| : \text{ every element has at most } \Delta \text{ neighbours, } \operatorname{diam}\Gamma(\mathcal{S}) \le D\}.$$

Distances $d(x, y)$ are taken in $\Gamma(\mathcal{S})$, and $\operatorname{diam}\mathcal{S}$ means $\operatorname{diam}\Gamma(\mathcal{S})$. Two distinct points on a common line are *collinear*; two distinct lines on a common point are *concurrent*. $\mathbb{F} = \mathrm{GF}(q)$ is the field of order q, and for q odd χ is its quadratic character, with $\chi(0) = 0$.

Lemma 2.1 (diameter in geometric terms)**.** *Let $\mathcal{S}$ have no isolated element and let $D \ge 2$.*

(i) If D is even, $\operatorname{diam}\mathcal{S} \le D$ *if and only if $d(p, L) \le D - 1$ for every anti-flag (p, L).*

(ii) If D is odd, $\operatorname{diam}\mathcal{S} \le D$ *if and only if any two points, and any two lines, are at distance at most* $D-1$.

In particular, $\operatorname{diam}\mathcal{S} \le 3$ *if and only if any two points are collinear and any two lines are concurrent, and* $\operatorname{diam}\mathcal{S} \le 4$ *if and only if for every anti-flag* (p, L) *some point of* L *is collinear with* p.

Proof. (i) Let $p \ne p'$ be points and L a line on p'. If $p \mathrel{I} L$ then $d(p,p') = 2$; otherwise $d(p,p') \le d(p,L)+1 \le D$. Lines likewise. (ii) For an anti-flag (p, L) take $p' \mathrel{I} L$; then $d(p,L) \le d(p,p')+1 \le D$. The converses hold by parity. □

Lemma 2.2 (adding flags, deleting twins)**.** *(a) Adding flags to an incidence structure keeps its incidence graph bipartite and does not increase any distance. (b) If two lines have the same set of points, deleting one of them does not change any distance between the remaining elements.*

Proof. (a) is clear. (b) A shortest path through the deleted line can use its twin instead; it does not already contain the twin, since then it could be shortened. □

2.2. **Generalized polygons.** A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; it has *order* (s,t) if every line has $s+1$ points and every point is on $t+1$ lines. It is *thick* if $s,t \ge 2$. A generalized 3-gon of order (q,q) is a projective plane of order q, with q^2+q+1 points and as many lines. A generalized quadrangle (GQ) of order (s,t) has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines, and its axiom is: for every anti-flag (p,L) exactly one point of L is collinear with p. A generalized hexagon of order (q,q) has $(q+1)(q^4+q^2+1)$ points and as many lines. By Feit and Higman [**?**], thick generalized n-gons exist only for $n \in \{2,3,4,6,8\}$, and a generalized octagon of order (s,s) does not exist; so $M_b(\Delta, D)$ is attained, for $\Delta \ge 3$, only at $D \in \{2,3,4,6\}$.

The projective plane. Points of $\mathrm{PG}(2,q)$ are written $(x : y : z)$ and lines $[u : v : w]$, with $(x:y:z) \mathrel{I} [u:v:w]$ if and only if $ux+vy+wz = 0$.

Quadrangles. $W(3,q)$ is the symplectic quadrangle: the points of $\mathrm{PG}(3,q)$ and the lines of $\mathrm{PG}(3,q)$ that are totally isotropic for a non-degenerate alternating form; it is a GQ of order (q,q) [**?**]. $Q(4,q)$ is the parabolic quadric of $\mathrm{PG}(4,q)$ with the lines it contains; it is a GQ of order (q,q), dual to $W(3,q)$ [**?**, 3.2.1]. The Ahrens–Szekeres quadrangle $\mathrm{AS}(q)$ (q odd) is introduced in coordinates in Section **??**.

The split Cayley hexagon. Let $V = \mathbb{F}^7$ with coordinates $(X_0, \dots, X_6)$,

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2, \qquad b(X,Y) = Q(X+Y) - Q(X) - Q(Y),$$

and let $Q(6,q)$ be the parabolic quadric $\{\langle x\rangle : Q(x) = 0\}$ of $\mathrm{PG}(6,q)$. Points $\langle x\rangle, \langle y\rangle$ of $Q(6,q)$ are *perpendicular*, $x \perp y$, if $b(x,y) = 0$, i.e. if the line xy lies on $Q(6,q)$. The *split Cayley hexagon* $H(q)$ [**?**], [**?**, §2.4] has as points all $N = N(q) = (q^6-1)/(q-1) = (q+1)(q^4+q^2+1)$ points of $Q(6,q)$, and as lines the lines of $Q(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ satisfy Tits' equations

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31}. \tag{2.1}$$

For example, the $q+1$ lines through $e_0 = (1,0,\dots,0)$ are $\langle e_0, \lambda e_5 + \mu e_6\rangle$, $(\lambda:\mu) \in \mathrm{PG}(1,q)$; they lie in the totally singular plane $\langle e_0, e_5, e_6\rangle$. The same equations define $H(q)$ for q even, with the quadric written $X_0X_4 + X_1X_5 + X_2X_6 + X_3^2$ (Section **??**).

The referee of the diameter-7 note recalls that this form of the quadric and of Tits' equations is [**?**, §2.4.13]. We could not consult [VM98] to confirm the place; please check and cite it.

Fact 2.3 ([**?**], [**?**, §2.4])**.** $H(q)$ is a generalized hexagon of order (q,q). The lines of $H(q)$ through a point x are the lines through x of a totally singular plane $\pi_x \subset x^\perp$. The collineation group contains $G_2(q)$; for $A \in \mathrm{SL}(3,q)$ the map $(v,a,w) \mapsto (Av, a, A^{-T}w)$, with $x=(v,a,w)$, $v=(x_0,x_1,x_2)$, $a=x_3$, $w=(x_4,x_5,x_6)$, is a collineation. $H(q)$ has a duality (an incidence-preserving bijection exchanging points and lines) if and only if q is a power of 3.

Let $I = I(q)$ be the incidence graph of $H(q)$, with $2N$ vertices, and $\Gamma_i(x) = \{y : d(x,y) = i\}$. Two elements are *opposite* if they are at distance 6, and $\mathrm{opp}(x) = \Gamma_6(x)$. The following facts are standard; we include the short proofs because Sections **??** and **??** use them repeatedly.

Lemma 2.4. *Let x, y be vertices of I.*

(a) If $d(x,y) \le 5$ there is a unique geodesic from x to y. If $1 \le d(x,y) = k \le 5$, exactly one neighbour of y is at distance $k-1$ from x and the other q at distance $k+1$; if $k=6$ all $q+1$ neighbours of y are at distance 5. Hence $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \le i \le 5$ and $|\Gamma_6(x)| = q^5$.

(b) (covering) *There is $z \in \Gamma_2(x)$ with $d(z,y) \le 3$ if x, y have different types, and with $d(z,y) \le 4$ if they have the same type.*

(c) (orthogonality) *For points x, y: $d(x,y) \le 4$ if and only if $x \perp y$. So $\mathrm{opp}(x) \cap \mathcal{P} = \mathcal{P} \setminus x^\perp$.*

Proof. (a) Two geodesics would close a cycle of length at most $10 < 12$. Neighbours of y are at distance $k \pm 1$ from x; the predecessor on the geodesic is at $k-1$, and a second one would give a second geodesic; at $k=6$ the diameter forces 5. The counts follow by induction. (b) If $2 \le d(x,y) \le 6$ take the vertex at distance 2 from x on a geodesic to y; if $d(x,y)=1$ take $z \in \Gamma_1(y) \setminus \{x\}$; if $x=y$ any $z \in \Gamma_2(x)$. The largest value of $d(z,y)$ is $6-2=4$ (same type) or $5-2=3$ (different types). (c) If $d(x,y) \le 4$ there is a point z with x, y on lines of $H(q)$ through z ($z=x$ if $d(x,y) \le 2$); these lines lie in π_z, so $b(x,y)=0$. Conversely $x^\perp \cap Q(6,q)$ is a cone with vertex x over a $Q(4,q)$ and has $1+q(q^3+q^2+q+1)$ points, which by (a) is the number $1+(q+1)q+(q+1)q^3$ of points at distance at most 4 from x. □

2.3. **Doubling with levels, and the hat lemma.** A *duplicate* of a vertex (a new vertex with the same neighbours) does not increase the diameter (see [**?**], which we have not seen, and [**?**, §7]). Delorme [**?**, §7] refined this into a doubling with levels, stated there without proof, and used it for the 2013 values at $D=4$ and $D=6$ [**?**, §8(a),(b)]. We state it for incidence structures and give a proof.

Let $\mathcal{A} = (\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \cdots \supseteq \mathcal{A}_r \supseteq \mathcal{A}_{r+1} = \emptyset)$ be a filtration of a set $\mathcal{A}_0$ of elements (points and lines may be mixed). The *level* of x is $\lambda(x) = \max\{k : x \in \mathcal{A}_k\}$, and $\lambda(x) = -1$ if $x \notin \mathcal{A}_0$. Fix $m \ge 2$. Two words over $[m] = \{1, \dots, m\}$ are *comparable* if one is a prefix of the other.

Definition 2.5 (Delorme)**.** The m-fold doubling $\mathcal{S}_{\mathcal{A}}^{(m)}$ has the *versions* x^w, for x an element of $\mathcal{S}$ and $w \in [m]^{\lambda(x)+1}$ (x^w is a point if x is a point and a line if x is a line), and

$$x^w \mathrel{\mathrm{I}} y^u \iff x \mathrel{\mathrm{I}} y \text{ and } w, u \text{ are comparable.}$$

So an element of level -1 has one version (identified with x), adjacent to every version of each of its neighbours; and a version x^w of level $k \ge 0$ is adjacent, for each neighbour y of x, to the single version $y^{w_1 \cdots w_{j+1}}$ if $\lambda(y) = j \le k$, and to the m^{j-k} versions y^{wu} ($|u| = j-k$) if $\lambda(y) = j > k$. This is the exact neighbourhood of every version. The map $x^w \mapsto x$ preserves incidence, so $d(x^w, y^u) \ge d(x,y)$.

The conditions on $\mathcal{A}$ are (with $D \ge 2$):

(L0) every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$;

(L1) for every $i \ge 0$: every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$;

(C_D) for any two elements x, y (possibly $x=y$) in the same connected component of the substructure induced on $\mathcal{A}_0$, there is $z \notin \mathcal{A}_0$ with $d(x,z) + d(z,y) \le D$.

Theorem 2.6 (Delorme [**?**, §7], (a)–(c))**.** *Let $\mathcal{S}$ have diameter at most $D \geq 2$ and let $m \geq 2$.*

(a) If (C_D) holds, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D.
(b) If (L0) and (L1) hold, every version satisfies $\deg x^w \leq \deg x + m - 1$, with equality only if x is incident with an element of level $\lambda(x) + 1$.
(c) $\mathcal{S}_{\mathcal{A}}^{(m)}$ has $|\mathcal{P}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{P}|$ points and $|\mathcal{L}| + (m-1)\sum_{k\geq 0} m^k |\mathcal{A}_k \cap \mathcal{L}|$ lines.
(d) Conversely, if $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most D, the conclusion of (C_D) holds for every pair $x, y \in \mathcal{A}_0$.

Delorme states (a)–(c) without proof ("je ne donne pas la preuve"); (d) is not in [**?**].

Proof. (c) An element of level k has $m^{k+1} = 1 + (m-1)(1 + m + \cdots + m^k)$ versions.

(b) Let $\lambda(x) = k$. By the description after Definition 2.5, only neighbours of level $> k$ contribute more than one version. If $k = -1$, (L0) gives at most one neighbour in $\mathcal{A}_0$; it has level exactly 0 (a neighbour in $\mathcal{A}_1$ would put x in $\mathcal{A}_0$ by (L1)), and contributes m versions. If $k \geq 0$, (L1) with $i = k$ gives at most one neighbour in $\mathcal{A}_{k+1}$, and (L1) with $i = k + 1$ gives none in $\mathcal{A}_{k+2}$; that neighbour contributes m versions. So $\deg x^w \leq \deg x - 1 + m$.

(a) For an infinite word ω the *sheet* Σ_ω is the set of versions x^w with w a prefix of ω. It contains one version of each element, and $x^w \mapsto x$ is an isomorphism from the substructure on Σ_ω onto $\mathcal{S}$. Two versions with comparable words lie in a common sheet, hence are at distance at most D. Let x^w, y^u have incomparable words; both are non-empty, so $x, y \in \mathcal{A}_0$. If x, y are in different components of the substructure on $\mathcal{A}_0$, a shortest path $x = z_0, \ldots, z_k = y$ ($k \leq D$) passes through some $z_i \notin \mathcal{A}_0$, whose unique version lies in every sheet. Lift $z_0, \ldots, z_i$ into a sheet through x^w and $z_i, \ldots, z_k$ into a sheet through y^u: this is a path of length k. If x, y are in the same component (including $x = y$), (C_D) gives $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \leq D$, and the same lifting gives a path of length at most D.

(d) Take versions x^w, y^u with w starting with the letter 1 and u with the letter 2, and a path of length at most D between them. Consecutive versions on it have comparable words, and comparable non-empty words start with the same letter; so some version on the path has the empty word, that is, it is an element $z \notin \mathcal{A}_0$. Projecting the path gives a walk $x \to z \to y$ of length at most D. □

Lemma 2.7 (hat lemma)**.** *Let $\mathcal{S}$ be a generalized n-gon of order (s, t), $s, t \geq 1$, and $\mathcal{A}_0$ a set of elements. Then (C_n) holds if and only if every connected component of the substructure induced on $\mathcal{A}_0$ is a tree of diameter at most $n - 2$.*

Proof. Suppose a component B contains a path $x = z_0, \ldots, z_{n-1} = y$ of length $n - 1$ (it does if B has diameter at least $n - 1$, and if B contains a cycle, since cycles have length at least $2n$). By the girth this is the only path of length at most $n - 1$ from x to y, so $d(x, y) = n - 1$. If $d(x, z) + d(z, y) \leq n$ then by parity $d(x, z) + d(z, y) = n - 1$, so z lies on a shortest x–y path, i.e. $z = z_i \in \mathcal{A}_0$. So ($C_n$) fails.

Conversely let B be a tree of diameter at most $n - 2$, $x, y \in B$, and P the path from x to y in B, of length d. The incidence graph is connected and has cycles, so some element of B has a neighbour outside $\mathcal{A}_0$. Let k be the least distance in B from P to such an element u, and z a neighbour of u outside $\mathcal{A}_0$. Then $d(x, z) + d(z, y) \leq d + 2k + 2$. If $k = 0$ this is at most n. If $k \geq 1$, every element of B at distance less than k from P has all its neighbours in B, and every element has at least two neighbours. So we can walk k steps away from P starting at x, and k steps away from P starting at y (along two different neighbours if $x = y$); in the tree B this gives a path of length $d + 2k$. Hence $d + 2k \leq n - 2$ and $d(x, z) + d(z, y) \leq n$. □

Corollary 2.8. *In a generalized n-gon of order (s, t), $s, t \geq 1$, for any filtration and $m \geq 2$, $\mathcal{S}_{\mathcal{A}}^{(m)}$ has diameter at most n if and only if every component of the substructure on $\mathcal{A}_0$ is a tree*

of diameter at most $n-2$. For $n=3$ the allowed components are single elements and flags; for $n=4$ also stars *(a point with some lines on it, or a line with some of its points); for $n=6$ trees of diameter at most 4.*

Proposition 2.9 (diameter 3 needs patches)**.** *In a projective plane, if $\mathcal{A}$ satisfies (L0), (L1) and (C_3), then $\mathcal{A}_0$ has at most one point and at most one line, and $\mathcal{A}_1 = \emptyset$.*

Proof. If $x \neq y$ are points of $\mathcal{A}_0$, the line xy is either outside $\mathcal{A}_0$ and incident with two elements of $\mathcal{A}_0$ (against (L0)), or in $\mathcal{A}_0$, and then x, xy, y is a path of length $2 > n-2$ in one component (against Lemma 2.7). Lines likewise. An element of $\mathcal{A}_1$ would have its $q+1 \geq 3$ neighbours in $\mathcal{A}_0$ by (L1). □

So pure doubling gains at most $2(m-1)$ elements in a projective plane; the hats of Section **??** double flags and then change a bounded number of incidences.

References